# Gautam Solar Private Limited

## IPQC Check Sheet

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| Document No. | GSPL/IPQC/IPC/003 |
| Issue Date | 01/12/2024 |
| Rev. No./Rev.Date | 01/30-08-2025 |

Date :- | Time :- | Shift | Po.no.:-

| Sr.N | Stage | Check point | Quantum of Check: Sample Size | Quantum of Check: Frequency | Acceptance Criteria | Monitoring Result | Remarks,if any |
|---|---|---|---|---|---|---|---|
| 1 | Shop Floor | Temperature | once | per shift | Temp. 25±3°C | Time 8:20Am 27.0C | |
| | | Humidity | once | per shift | RH ≤60% | Time 8:20Am 33°C | |
| 2 | Glass Loader | Glass dimension(L* | once | per shift | As Per PO | 2376*1128*2.0mm | |
| | | Appearance(Visual) | once | per shift | Glass Broken, Crack, Scratches | OK | |
| 3 | EVA/EPE Cutt | EVA/EPE Type | once | per shift | As per approved BOM | | |
| | | EVA/EPE dimension( | once | per shift | As per Specification | | |
| | | EVA/EPE Status | once | per shift | Not allowed dust & foreign part | | |
| 4 | Eva/EPE Sold | Soldering Tempratu | Once | per shift | As per specification and Should | 385C | |
| 5 | Cell Loading | Cell Manufacturer | once | per shift | Refer Process Card | | |
| | | Cell Size(L*W) | once | per shift | Refer Process Card | | |
| | | Cell Condition | once | per shift | Free From dust,finger spot,color var | | |
| | | Cleanliness of Cel | once | per shift | No unwanted or waste material s | | |
| | | Verification of Pr | once | per shift | ATW Stringer Specificatio | Monitoring of ATW STRINGER | |
| | | Cell Cross cutting | once | per shift | Both side cutting should be equ | | |
| 6 | Tabber & str | Verification of Pr | once | Month | ATW Stringer specificatio | ATW Temprature Validation | |
| | | Visual Check after | once | 1 String/TS | TS Visual Criteria | TS01A, TS01B, TS02A, TS02B, TS03A, TS03B, TS04A, TS04B | |
| | | EL Image of String | once | 1 String/TS/ | TS EL Criteria | TS01A, TS01B, TS02A, TS02B, TS03A, TS03B, TS04A, TS04B | |
| | | String length | once | 1 String/Str | Refer Process Card | TS01A, TS01B, TS02A, TS02B, TS03A, TS03B, TS04A, TS04B | |
| | | Cell to Cell Gap | | | | TS01A, TS01B, TS02A, TS02B, TS03A, TS03B, TS04A, TS04B | |

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|---|---|---|---|---|---|---|---|---|---|---|---|
| | | Verification of Sold | 2 cell eac | per shift | Peel Strength ≥1N | Ribbon to cell soldering peel strength | | | | | |
| 7 | Auto bussing | String to String Gap | once | per shift | Refer Process Card & Module Drawing | | | | | | |
| | | Cell edge to Glass e | once | per shift | | TOP | | | | | |
| | | | | | | Bottom | | | | | |
| | | | | | | Sides | | | | | |
| | | Soldering Peel Stren | once | per shift | ≥2N | Ribbon to busbar peel test | | | | | |
| | | Terminal busbar to e | once | per shift | 132 Cell module drawing | | | | | | |
| | | | | | Refer Module Drawing: GSPL/ | | | | | | |
| | | Soldering Quality of | Every 4h | per shift | No Dry/Poor Soldering | | | | | | |
| | | Top & Bottom Creepag | Every 4h | per shift | Creepage distance should be as per | | | | | | |
| | | Verification of Proc | once | per shift | Specification for Auto Buss | | | | | | |
| | | Quality of auto tapi | Every 4h | per shift | Taping should be proper,no Cell Shi | | | | | | |
| 8 | Auto RFID Lo | Position verificatio | Every 4h | per shift | Should not be tilt | | | | | | |
| 9 | EVA/EPE cutti | EVA/EPE Type | once | per shift | EVA | | | | | | |
| | | EVA/EPE dimension(L* | once | per shift | As per Specification | | | | | | |
| | | EVA/EPE Status | once | per shift | Not allowed dust & foreign particle | | | | | | |
| 10 | Back Glass Lo | Glass dimension(L*W* | once | Per shift | As per PO | | | | | | |
| | | No. of Holes/ Holes | once | Per shift | 3 hole with dimension 12mm±0.5mm | | | | | | |
| 11 | Auto Busbar F | Visual Inspection | 5 pieces | per shift | No cracks/ breaks in busbar &prope | | | | | | |

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|---|---|---|---|---|---|---|---|---|
| 12 | Pre lamina | EL Inspection a | 5 piece | per shift | Pre EL Inspection Cr | S.No | GS04890TCS+00231567 | |
| | | | | | | S.No | GS04890TCS+00231568 | |
| | | | | | | S.No | | |
| | | | | | Pre EL Visual Criter | S.No | | |
| | | | | | | S.No | | |
| 13 | String Rew | cleaning of rew | once | per shift | Rework Station should be C | | | |
| | | Soldering Iron | once | per shift | 400±30°C | Time: | | |
| 14 | Module Rew | Method of Rewor | once | per shift | As per WI (GSPL/P/WI/012) | | | |
| | | Cleaning of Rew | once | per shift | Rework Station should be C | | | |
| | | Soldering Iron | once | per shift | 400±30°C | Time: | | |
| 15 | Laminator | Monitoring of L | once | per shift | Process Parameter of | | | |
| | | Cleaning of Dia | once | 24h | Diaphragm/Release sheet sh | | | |
| | | Peel of Test b/ | All pos | All lamina | E/G ≥60N/cm E/B≥60N/cm | Refer Document GSPL/IPQC/QC/001 | | |
| | | Gel Content Tes | | | 75to 95% | Refer Document GSPL/IPQC/QC/001 | | |

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|---|---|---|---|---|---|---|---|---|---|---|---|
| 16 | Auto Tap | Visual Check | 5 piec | per shif | Check Tape Removing | S | | | | | |
| 17 | Auto Edg | Trimming Qua | 5 piec | per shif | Excess layer from the | S.No. | | | | | |
| | | | | | | S.No. | | | | | |
| | | | | | | S.No. | | | | | |
| | | | | | | S.No. | | | | | |
| | | | | | | S.No. | | | | | |
| | | Trimming Bla | once | per mont | Worn out not allowed | | | | | | |
| 18 | 90° Visu | Visual Inspe | 5 piec | per shif | Post Lam Visual I | S.No | | | | | |
| | | | | | | S.No | | | | | |
| | | | | | | S.No | | | | | |
| | | | | | | S.No | | | | | |
| | | | | | | S.No | | | | | |
| 19 | Framing | Glue uniform | 1 set | per shif | Should be uniform,Bac | | | | | | |
| | | Short Side G | once | Per shif | Fill as per Specifica | 22.113GM | | | | | |
| | | Long Side Gl | once | Per shif | | | | | | | |
| | | Anodizing Thi | once | Per shif | ≥15 micron | 19.5 Micron | | | | | |

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|---|---|---|---|---|---|---|---|---|---|---|---|---|
| 20 | Junction Bo | Junction Box(Conn | once | Per shift | As per Process Card & module | | | | | | | |
| | | Silicon Glue Weig | once | Per shift | 21±6 gm | | | | | | | |
| 21 | Auto JB Sol | Max Welding time | once | Per shift | As per Specification | | | | | | | |
| | | Soldering current | once | per shift | As per Specification | | | | | | | |
| | | Soldering Quality | once | per shift | Welding area should be fully | | | | | | | |
| 22 | JB Potting | A/B Glue Ratio | once | Per shift | As per Specification | Refer Document GSPL/IPQC/QC/011 | | | | | | |
| | | Potting material | once | Per shift | 21±6 gm | | | | | | | |
| | | Nozzle Changing | once | every 6h | Should be changed after 6 hou | Time: | | | Time | | | |
| 23 | OLE Potting | Visual Check | once | 5 piece | Potting should be properly fi | | | | | | | |
| 24 | Curing | Temperature | once | per shift | 25±3■ | | | | | | | |
| | | Humidity | once | per shift | ≥50% | | | | | | | |
| | | Curing Time(h) | once | Per shift | ≥4 hours | | | | | | | |
| 25 | Buffing | Corner Edge,Buffi | 5 pieces | per shift | Should not be sharp & No worn | | | | | | | |
| 26 | Cleaning | Module should be | 5 pieces | per shift | Post Lam Visual Cr | S.No | | | | | | |
| | | | | | | S.No | | | | | | |
| | | | | | | S.No | | | | | | |
| | | | | | | S.No | | | | | | |
| | | | | | | S.No | | | | | | |

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|---|---|---|---|---|---|---|---|---|---|---|
| 27 | Flash Test | Ambient Temp. | once | per shift | 25±3■ | | | | | |
| | | Module Temp. | once | per shift | 25±3■ | | | | | |
| | | Sunsimulator Cal | once | 12h | Sunsimulator must be calibr | | | | | |
| | | Validation | once | every 6h | As per GSPL/QA/S/11 | | | | | |
| | | Silver Reference | once | Two weeks | Should be same as orignal E | | | | | |
| 28 | Hipot Test | DCW/IR/Ground C | 5 pieces | per shift | ≤50µA , >40MΩ.m2 ,(0-100) m | S.No | DCW | IR | Ground | |
| | | | | | | | | | | |
| | | | | | | | | | | |
| | | | | | | | | | | |
| | | | | | | | | | | |
| | | | | | | | | | | |
| 29 | Post EL Te | Voltage & Curren | once | Shift | As per WI (GSPL/P/WI/027) | | | | | |
| | | EL Inspection an | 5 pieces | per shift | Post EL Inspection Cr | S.No | | | | |
| | | | | | | S.No | | | | |
| | | | | | | S.No | | | | |
| | | | | | Post EL Visual Criter | S.No | | | | |
| | | | | | | S.No | | | | |
| 30 | RFID | RFID Position | once | per shift | As per Process card | | | | | |
| | | Cell & Module Ma | once | per shift | As per BOM and Process card | | | | | |

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|---|---|---|---|---|---|---|---|---|
| 31 | Final Visua | Visual Inspection | 5 pieces | per shift | Post lam visual inspect | S.No. | | |
| | | | | | | S.No. | | |
| | | | | | | S.No. | | |
| | | | | | | S.No. | | |
| | | | | | | S.No. | | |
| | | Backlabel | 5 pieces | per shift | Air bubble,Tilt,Misprint,folde | S.No | | |
| | | | | | | S.No | | |
| | | | | | | S.No | | |
| | | | | | | S.No | | |
| | | | | | | S.No | | |
| 32 | Dimension m | L*W and Module Pr | once | per shift | As per Module drawing (±1mm) | | | |
| | | Mounting Hole X & | once | Per shift | | | | |
| | | Diagonal Differen | once | Per shift | ≤3mm | | | |
| | | Corner Gap | once | Per shift | As per visual Inspection crite | | | |
| | | JB Cable length | once | Per shift | As per Process card | | | |
| 33 | Packaging | Packaging Label | once | Per shift | WI For Packaging | | | |
| | | Content in Box | once | Per shift | | | | |
| | | Box Condition | once | Per shift | Damage,dull printing,wet boxes | | | |
| | | Wooden Pallet dim | once | Per shift | Should not be less than module | | | |
| Checked by | | | | | | Reviewed By | | |